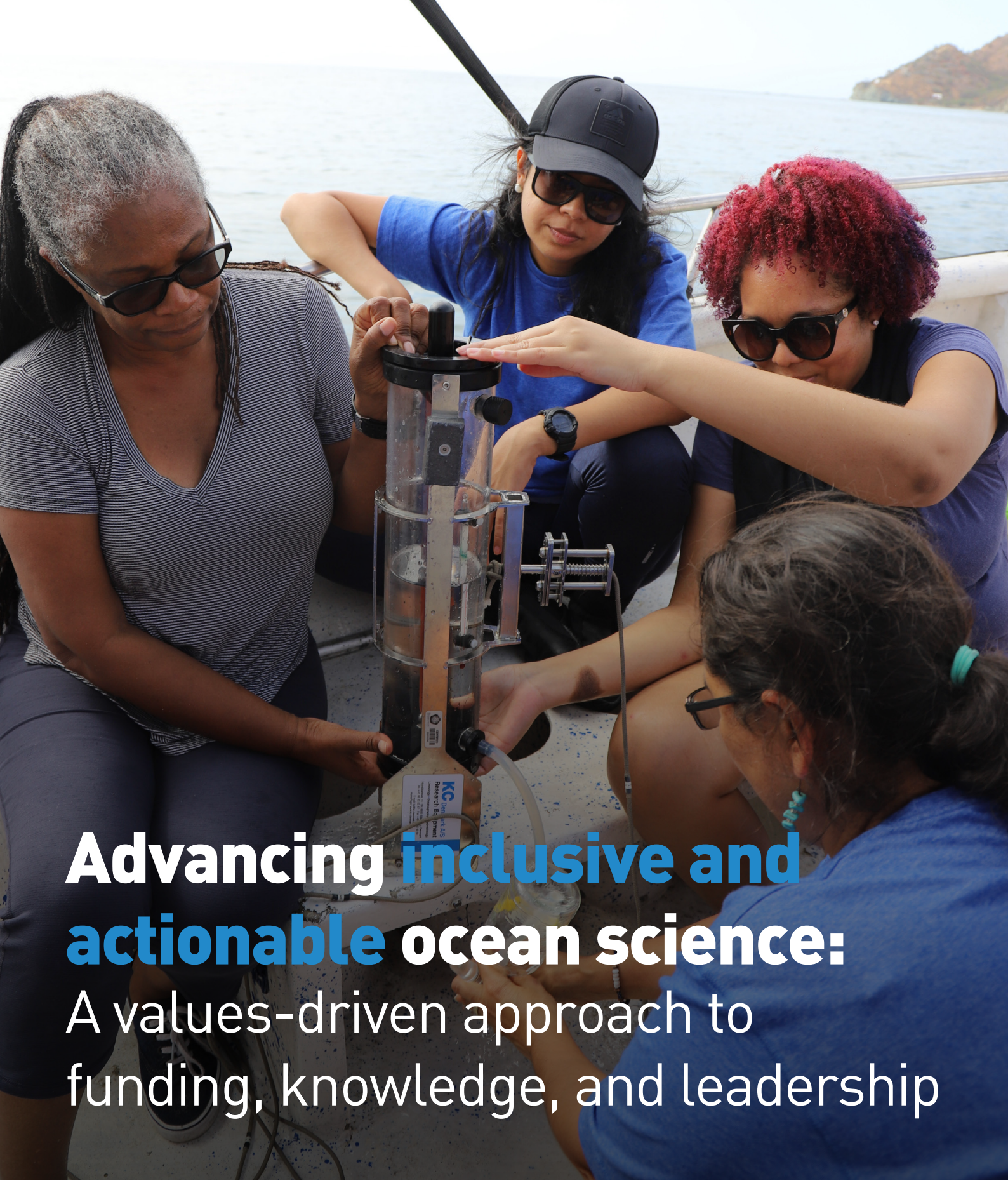

Advancing inclusive and actionable ocean science:

A values-driven approach to funding, knowledge, and leadership

The United Nations
Decade of Ocean Science
For Sustainable Development
2021-2030

2021 2030 United Nations Decade of Ocean Science for Sustainable Development

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1 Introduction

The United Nations (UN) Decade of Ocean Science for Sustainable Development 2021-2030 (Ocean Decade) calls for a revolution in how ocean science is produced, utilized, and disseminated.[1] To advance this transformation, the Intergovernmental Oceanographic Commission (IOC) of UNESCO convened more than 1,500 in-person participants and thousands virtually for the 2024 UN Ocean Decade Conference to discuss the development of a common vision for the future priorities of the Ocean Decade and the broader global ocean science agenda.[2] The resulting Barcelona Statement identified key priorities for ocean science across ocean observations, marine pollution monitoring, and interoperable ocean data. Furthermore, it highlighted the need for the enhanced use of innovative technologies to enable the equitable generation, access, and use of observations, data, and knowledge for all stakeholders.[3] Finally, the Barcelona Statement emphasized the critical need to address cross-cutting issues to achieve these ambitions, including improving the recognition of all knowledge systems, increasing and expanding financing for ocean science, and enhancing diversity, equity, and inclusivity in the Ocean Decade, while systematically removing barriers to generational, geographic, and gender diversity.[4] In short, there is a profound need for inclusive and actionable ocean science that is co-designed with those on the frontlines of ocean change and responds directly to their priorities, producing knowledge that informs policy and decision-making and addresses the numerous drivers of ocean decline.

As a result, the Foundations Dialogue, an informal, global network of community, corporate, and private philanthropic funders working together to support the Ocean Decade, understood that values underpin the delivery of inclusive and actionable ocean science. Furthermore, inclusive and actionable ocean science is the essential link to producing the knowledge and scientific data needed for evidence-based decision-making on the broad range of policies and investments required to fulfill the ambitions of the 2030 Agenda for Sustainable Development and multilateral governance frameworks. In response to the Barcelona Statement, Members convened in Rio de Janeiro, Brazil, for the Fourth Foundations Dialogue in September 2024 to discuss the need for mobilizing funding to support the Ocean Decade and how to ensure its equitable distribution. The Rio Action Statement affirmed the need for improved practices and committed to developing a position paper on the values that support inclusive and actionable ocean science.[5] At the Fifth Foundations Dialogue in Tokyo, Japan, Members reaffirmed their commitment to these principles and emphasized the need to strengthen contributions to multilateral frameworks and build resilient, future-proof ocean science systems.[6]

Values are not new, but this is a new moment. With five years remaining in the Ocean Decade, there is a powerful opportunity to galvanize global actors to address ocean decline and support the ambitions of the 2030 Agenda for Sustainable Development. The Foundations Dialogue understands the vital role that funders play in shaping inclusive and actionable ocean science. While funders are not solely responsible for the systemic barriers inherent in ocean science, discussed below, they are not neutral actors. In the context of philanthropy, values are embedded

[1] Intergovernmental Oceanographic Commission, *UN Ocean Decade Implementation Plan*. Paris: UNESCO, 2020.
[2] Intergovernmental Oceanographic Commission, *Barcelona Statement of the 2024 Ocean Decade Conference*. Paris: UNESCO, 2024.
[3] Ibid.
[4] Ibid.
[5] Foundations Dialogue of the Ocean Decade, *Rio Action Statement of the Fourth Foundations Dialogue Meeting*. 2024.
[6] Foundations Dialogue of the Ocean Decade, *Tokyo Statement of the Fifth Foundations Dialogue Meeting*. 2025.

in all decisions, from who designs a call for proposals to how grantee impact is measured to which knowledge systems are recognized as legitimate. However, these values often remain implicit, unexamined, or filtered through institutional constraints. This is not simply a call for more ocean science; it is a call to transform how ocean science is imagined, funded, and practiced, so that it reflects the values, priorities, and leadership of those impacted by ocean change.[7]

This effort by the Foundations Dialogue comes at a moment of increased multilateral activity. In June 2025, the United Nations Ocean Conference (UNOC) brought together leaders from the UN, government, business, academia, civil society, philanthropy, and the scientific community, as well as Indigenous Peoples, youth, and local communities, to discuss the urgent need to accelerate action to conserve and sustainably use the ocean, seas, and marine resources.[8]

At UNOC, delegates adopted "Our ocean, our future: United for urgent action" (Nice Ocean Action Plan), which acknowledged that action is not advancing at the speed or scale necessary to achieve Sustainable Development Goal (SDG) 14 and the broader ambitions of the 2030 Agenda for Sustainable Development.[9] Delegates called for just, transformative, and inclusive efforts that address the many ongoing threats to ocean health and empower all stakeholders, including women and girls, Indigenous Peoples, youth, local communities, and those in vulnerable situations, to be included in decision-making.[10]

The Nice Ocean Action Plan also affirmed the importance of the Ocean Decade in transforming the way science is conducted in service of a healthy, equitable ocean.[11] That ocean science is essential to addressing ocean health and unlocking opportunities to achieve the development agenda is not a new revelation for the many stakeholders engaged in the Ocean Decade. For the last five years, the Ocean Decade has provided a galvanizing framework that has endorsed more than 60 global programmes and over 500 projects to help generate "the science we need for the ocean we want."[12]

While ocean knowledge is expanding, significant gaps still remain in seabed mapping, species identification, and ocean observation.[13] These and other gaps limit the ability to respond to accelerating ocean change and provide the necessary knowledge and data for evidence-based decision-making at the local level and, more broadly, to support the momentum of multilateral governance frameworks, including the Agreement under the UN Convention on the Law of the Sea on the Conservation and Sustainable Use of Marine Biological Diversity of Areas beyond National Jurisdiction (BBNJ Agreement), the Kunming-Montreal Biodiversity Framework, the UN Plastics Treaty, the UN Convention to Combat Desertification (UNCCD), or the UN Framework Convention on Climate Change (UNFCCC) agenda.

Now, at the midpoint of the Ocean Decade, the stakes are higher than ever. Changes must be made to include those on the frontlines of ocean change, including Indigenous knowledge holders, Small Island Developing States (SIDS), coastal communities, and early-career professionals to ensure

[7] Personal conversation with Dr. Rachel Clausing, Special Advisor, Fondation Prince Albert II de Monaco.
[8] United Nations, *Our Ocean, Our Future: United for Urgent Action – Outcome Document of the 2025 UN Ocean Conference.* 2025.
[9] Ibid.
[10] Ibid.
[11] Ibid.
[12] Vidar Helgesen, Remarks at the Ocean Decade Forum, Executive Secretary, IOC, 11 June 2025.
[13] Ibid.

they are not simply engaged but allowed to lead. At the same time, progress must be made to close the time gap between knowledge generation and its application through policymaking.[14]

To help close these gaps, the Ocean Decade has advanced co-design, a multi-stage process that enables transformative solutions to change the trajectory of ocean science.[15] Central to co-design is the necessity of delivering science that is co-identified, co-produced, and co-delivered in a multi-stakeholder, transdisciplinary environment that incorporates local and Indigenous knowledge, ensuring it remains relevant and responsive throughout the entire value chain,[16] from knowledge generation to applications, services, shared benefits, and the utilization of science for solutions.[17] To be successful, there must be an intentional application of values, including justice, equity, diversity, inclusion, pluralism, and others, to ensure that rhetoric aligns with action and avoids values-washing.[18] This intentional application aligns with the commitments expressed in the Barcelona Statement to enhance the recognition of Indigenous and local knowledge, continue enhancing diversity, inclusivity, and equity within the Ocean Decade, and address ongoing barriers to generational, geographic, and gender diversity.[19]

Yet ambition alone will not deliver transformation. The systems that govern ocean science still mirror historical barriers and inequities. Power asymmetries persist across funding, authorship, and governance, where decision-making authority and control remain in high-income countries.[20] Parachute science and tokenistic partnerships, epistemic injustice, geographic and linguistic barriers, exclusion from global policy spaces, or inequitable data practices remain.[21] Moreover, while there is recognition that diverse, participatory, and co-produced science consistently leads to more innovative, trusted, and applied outcomes, barriers still exclude Global South researchers, Indigenous knowledge holders, women, disabled scientists, and early-career professionals.[22] The barriers are not just technical. They are structural, and they reflect values, implicit or explicit, about whose knowledge matters, who sets the agenda, and who benefits from research. Business-as-usual approaches, dressed in transformative language, will not close the necessary gaps in inclusive and actionable ocean science.

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[14] Ibid.

[15] Alice B.M. Vadrot et al., "Towards a Reflexive, Policy-Relevant and Engaged Ocean Science for the UN Decade: A Social Science Research Agenda," *Earth System Governance* 14 (2022): 100150. https://doi.org/10.1016/j.esg.2022.100150.

[16] Personal conversation with Dr. Rachel Clausing, Fondation Prince Albert II de Monaco.

[17] Intergovernmental Oceanographic Commission, *Co-Designing the Science We Need for the Ocean We Want*. Paris: UNESCO, 2021.

[18] Personal conversation with Professor Heather Koldewey, Lead, Bertarelli Foundation's Marine Science Programme, Zoological Society of London (ZSL).

[19] Intergovernmental Oceanographic Commission, *Barcelona Statement of the 2024 Ocean Decade Conference.* Paris: UNESCO, 2024.

[20] Pavel Tolochko and Alice B.M. Vadrot, "The Usual Suspects? Distribution of Collaboration Capital in Marine Biodiversity Research," *Marine Policy* 124 (2021): 104318. https://doi.org/10.1016/j.marpol.2020.104318.

[21] Asha de Vos et al., "Towards Equity and Justice in Ocean Sciences," *NPJ Ocean Sustainability* 2, no. 25 (2023). https://doi.org/10.1038/s44183-023-00028-4.

[22] Ana K. Spalding et al., "Engaging the Tropical Majority to Make Ocean Governance and Science More Equitable and Effective," *NPJ Ocean Sustainability* 2, no. 25 (2023). https://doi.org/10.1038/s44183-023-00015-9.

2 This position paper

Much has been written about the need to increase funding for ocean science and how SDG 14 remains the most underfunded of the SDGs. Furthermore, philanthropic funding for ocean science and conservation doubled to US$1 billion in 2022 from 2010, but still accounted for less than 1% of the global philanthropic spending in 2022.[23] While this is a critical issue, this position paper will not address the funding gaps. The recent IOC position paper, "*Benefits of investment in ocean science: Developing an evidence base to guide decision-making*," covers these issues in detail and should be considered the primary resource for this topic.

This position paper aims to stimulate dialogue on how values underpin the actions taken by funders that shape inclusive and actionable ocean science, leading to evidence-based decision-making at local, national, regional, and global levels. Members of the Foundations Dialogue understand that this exercise is not about declaring the one set of values that everyone should adopt. Instead, this position paper highlights what funders are already learning through their experiences, and as a result, what works, what does not, and where values emerge as critical levers in the pursuit of inclusive and actionable science.[24] Members of the Foundations Dialogue have also committed to sharing this position paper within their peer networks to stimulate discussion and share best practices to support inclusive and actionable ocean science. This work is intended as the basis for broader action. The Foundations Dialogue also aims to develop a living set of guidelines for grantmakers and grantees on how to act on these values and provide associated skills development.[25]

This position paper draws on the extensive literature on barriers to inclusive and actionable ocean science, as well as interviews with Members of the Foundations Dialogue and those impacted by these barriers, including early-career professionals, Indigenous knowledge holders, coastal communities, and small island nations. Case studies presented in this position paper aim to highlight actions and activities that support a values-based approach to inclusive and actionable ocean science, and do not represent the only initiatives underway.

As noted above, while this position paper does not focus on funding gaps, the Members of the Foundations Dialogue acknowledge that values-based practices can be significantly strengthened by innovative financial mechanisms. Blended finance, adaptive and iterative grantmaking, and flexible multi-year and multilateral structures are emerging as powerful enablers to operationalize values such as equity, inclusivity, and pluralism. By combining philanthropic, public, and private resources, these mechanisms not only expand funding pools but also ensure that financial flows align with the transformative ambition of inclusive and actionable ocean science.[26]

[23] Intergovernmental Oceanographic Commission, *Benefits of Investment in Ocean Science: Developing an Evidence Base to Guide Decision-Making*. Paris: UNESCO, 2025.
[24] Personal conversation with Dr. Jason Landrum, Program Director, Blue Convergence Fund.
[25] Foundations Dialogue of the Ocean Decade, *Rio Action Statement of the Fourth Foundations Dialogue Meeting*. 2024.
[26] Elizabeth H. Stephenson et al., "Small grants advance global ocean conservation and management equity," *Biological Conservation* 300 (2024): 110845. https://doi.org/10.1016/j.biocon.2024.110845.

3 A theory of change: Centering values for inclusive and actionable ocean science

Members of the Foundations Dialogue recognize that the ambition of inclusive and actionable ocean science aligns with the objectives of the 2030 Agenda for Sustainable Development and multilateral governance frameworks. Further, Members of the Foundations Dialogue believe that evidence-based decision-making must underpin the policies and investments that are core to achieving the ambitions of these frameworks. In this context, evidence-based decision-making draws on ocean science that is:

- Designed and implemented through a collaborative, inclusive, and transparent process;
- Respectful of and synthesizes the best available, plural knowledge from transdisciplinary stakeholders to conserve, protect, and sustainably use marine ecosystems and resources;
- Structured to fairly and equitably distribute benefits and risks.

To achieve this requires urgent change at scale to address the barriers noted above and described in this position paper, which roots power, funding, and decision-making authority in the Global North,[27] thereby sidelining many stakeholders from agenda-setting, leadership, and benefit-sharing. These inequities are compounded by the inconsistent or superficial use of co-design approaches, which undermines both the legitimacy and usability of science, particularly for the communities most affected by ocean change.[28] Achieving systemic transformation will also require aligning scientific ambition with finance reforms and global policy agendas that can sustain inclusive and actionable approaches beyond individual projects or funding cycles. Funders have an outsized influence in reshaping not just what science is conducted, but also how this science is carried out and by whom. The midpoint of the Ocean Decade creates a unique opportunity for bold shifts in who leads, what knowledge is legitimized, how research is funded, and who benefits from ocean science.

Accordingly, this Theory of Change centers on enabling inclusive and actionable ocean science that is designed with and for those affected to ensure that knowledge and data are funded, generated, shared, and applied to deliver:

Expected outcomes

- A more equitable ocean science ecosystem where power, credit, and benefits are redistributed.
- Greater trust, relevance, and usability of ocean knowledge, grounded in the lived realities and needs of diverse knowledge holders and users.
- Stronger uptake and legitimacy of ocean knowledge in policy and finance, through alignment with community priorities and co-designed processes.
- Evidence-based decision-making becomes routine in policy and investment decisions for the conservation, protection, and sustainable use of marine ecosystems and resources.
- Enhanced measurement systems to transparently track how values-based practices lead to measurable changes in both outputs (e.g. co-designed report, funding call explicitly shaped by multiple knowledge systems, etc.) and outcomes (e.g. inclusion, policy uptake, evidence-based decision-making, and knowledge-sharing).[29]

[27] Tolochko et al., "The Usual Suspects? Distribution of Collaboration Capital in Marine Biodiversity Research," *Marine Policy.*
[28] Stephenson et al., "Small grants advance global ocean conservation and management equity," *Biological Conservation.*
[29] Personal conversation with Janaína Bumbeer, Project Manager, Fundação Grupo Boticário de Proteção à Natureza.

To deliver these outcomes consistently and at scale, this Theory of Change is grounded in the following working assumptions about how ocean science is funded, generated, governed, and used:

Key assumptions

- Systemic barriers that impede inclusive and actionable ocean science persist and must be explicitly addressed.
- Funders have unique leverage to influence research norms, institutional incentives, and ecosystem-wide practices.
- Diverse knowledge systems improve the legitimacy, quality, credibility, and relevance of scientific data.
- Co-design is a proven tool for achieving inclusive and actionable ocean science, particularly when authentically applied throughout the research lifecycle.
- Data gathering must consider rights and follow the principles of Free, Prior, and Informed Consent (FPIC) and relevant governance protocols, including CARE and FAIR[30] practices where appropriate.
- Values-based ocean science cannot be scaled or sustained without explicit connections to financial and governance systems, and alignment ensures that knowledge is not only produced inclusively but also embedded within durable structures that sustain its application and relevance.

These key assumptions reveal what leverage funders and other actors in the ocean science community have in order to change systems, practices, and protocols:

Strategic levers for change

1. Shift who leads and who benefits

Fund diverse leadership, redistribute decision-making power, and ensure co-sharing of science, credit, and outcomes.

2. Redefine what counts as knowledge

Elevate co-designed research, plural knowledge systems, transdisciplinary perspectives, and long-term relationships as key criteria for grantmaking.

3. Incentivize values-based practices

Tie funding and institutional support to demonstrated action on values, including justice, equity, diversity, inclusion, and pluralism, that responds to knowledge holders' and users' priorities.

Operationalizing these strategic levers into clear, repeatable practices requires the development and implementation of pathways of action that convert intent into norms:

[30] CARE: Collective benefit, Authority to control, Responsibility, and Ethics. FAIR: Findable, Accessible, Interoperable, and Reusable.

Pathways of action

- **For funders:**

Adopt values-based criteria in funding calls;[31] require authentic co-design across planning, budgeting, and authorship; invest in underrepresented institutions and regional networks; support multilingual dissemination;[32] and hold grantees accountable for inclusive practices.[33]

- **For researchers and institutions:**

Center co-design from the outset of research; integrate transformational and relational values across team structures, community engagement, data governance, knowledge sharing, and benefits.

- **For multilateral and policy forums:**

Advocate for procedural justice in science-policy interfaces and support mechanisms that elevate underrepresented voices in global governance.

Embedded within each of these assumptions, levers, and pathways are behaviors and decisions guided by values that serve as mechanisms for change. Values such as justice, equity, diversity, inclusion, and pluralism are not aspirational but are core drivers of transformation. However, to achieve the desired ambitions now and in the future, values must be operationalized, not merely named, to effect meaningful shifts in outcomes.[34]

[31] Stephenson et al., "Small grants advance global ocean conservation and management equity," *Biological Conservation*.
[32] Personal conversation with Janaína Bumbeer, Fundação Grupo Boticário de Proteção à Natureza.
[33] Tolochko et al., "The Usual Suspects? Distribution of Collaboration Capital in Marine Biodiversity Research," *Marine Policy*.
[34] Personal conversation with Professor Heather Koldewey, ZSL.

4 The UN Decade of Ocean Science for Sustainable Development (2021-2030)

Launched in 2021 through a resolution by the UN General Assembly, the UN Decade of Ocean Science for Sustainable Development 2021-2030, or Ocean Decade, provides a convening framework for a wide range of stakeholders worldwide to engage and collaborate beyond their traditional communities, triggering a revolution in ocean science.[35]

Through the Ocean Decade, stakeholders generate the science and knowledge needed to address the ten Ocean Decade Challenges and provide the insights required to shape global, regional, and national policies, aligning with the 2030 Agenda for Sustainable Development and the SDGs, as well as other governance frameworks.[36] By collectively aligning research, investments, and initiatives around a set of common challenges, the Ocean Decade community will contribute to a well-functioning, productive, resilient, sustainable, and inspiring ocean.[37]

Now at its midpoint, the Ocean Decade has strengthened the ability to galvanize actors through operating structures that enable collaboration globally, including the Decade Alliance, the Decade Coordination Unit, Collaborative Centers, Coordination Offices, and Implementing Partners, as well as regional task forces, National Decade Committees, and Communities of Practice on thematic action areas such as Africa, coastal ecosystem and community resilience, ocean-climate nexus, ocean observations, and more. To date, the Ocean Decade has endorsed over 60 global programmes and more than 550 projects to support robust, accessible, and timely ocean knowledge.

Understanding that the ocean is changing rapidly, with profound consequences for climate regulation, biodiversity, food security, and livelihoods, the importance of the Ocean Decade cannot be overstated. The focus for the remaining five years is to urgently scale efforts to support the achievement of the development agenda and build on the momentum of multilateral governance frameworks. As such, the Ocean Decade will deepen its emphasis on co-design and prioritize the inclusion of frontline communities to unlock the transformative potential of science conducted not just for society, but with society.

[35] Intergovernmental Oceanographic Commission, *Ocean Decade Website.* Paris: UNESCO, 2025. https://oceandecade.org.
[36] Intergovernmental Oceanographic Commission, *Ocean Decade Progress Report 2023–2024.* Paris: UNESCO, 2025.
[37] Intergovernmental Oceanographic Commission, *Ocean Decade Website.*

5 The Foundations Dialogue of the Ocean Decade

Launched in 2020, the Foundations Dialogue is an informal network of more than 25 philanthropic organizations working collaboratively to support the Ocean Decade. Members of the Foundations Dialogue understand the critical role that philanthropic funders can play in addressing the many issues that compromise ocean health and resilience.[38] Furthermore, philanthropic organizations can serve as conveners, invest in innovative approaches, and catalyze collaborative efforts with a diverse range of stakeholders.[39]

Since 2020, the Foundations Dialogue has operated through a combination of virtual and in-person meetings to deepen commitment to developing and resourcing endorsed Decade Actions, sharing best practices, and advancing philanthropic engagement with the Ocean Decade. Discussions and outcomes of the Foundations Dialogue meetings have resulted in the development of the 2022 Bouknadel and 2023 Monaco Statements. At the Fourth Foundations Dialogue Meeting in Rio de Janeiro, Brazil, hosted by the Boticario Foundation, Members recognized that achieving the ambitions of the Ocean Decade required not just a substantial increase in financial resources, but also better practices for investing in inclusive and actionable ocean science.[40] As a result, Members committed to developing this position paper and a living set of guidelines for grantmakers and grantees to improve values-based actions within ocean science.

Fourth Foundations Dialogue Meeting in Rio de Janeiro, Brazil. © Carol Machado / Fundação Grupo Boticário*

[38] Intergovernmental Oceanographic Commission, *Ocean Decade Website*.
[39] Ibid.
[40] Foundations Dialogue of the Ocean Decade, *Rio Action Statement of the Fourth Foundations Dialogue Meeting.* 2024.

6 Defining inclusive and actionable ocean science

The Ocean Decade has called for a revolution in how science is conducted, shared, and applied. Central to this transformation is the understanding that ocean science must be both inclusive and actionable. For the purpose of this position paper, we define inclusive and actionable ocean science as follows:

Inclusive ocean science: Represents the intentional identification, design, and implementation of ocean research that equitably and respectfully involves diverse stakeholders across peoples, places, knowledge systems, disciplines, and sectors, particularly those who have been historically excluded or marginalized. In practice, this means:

- Representation: Ensuring that women and youth, early-career professionals as well as researchers from the Global South, Black, Indigenous, and People of Color (BIPOC), Lesbian, Gay, Bisexual, Transgender, Queer, Intersex and Asexual (LGBTQIA+), and persons with disabilities are engaged as leaders, not just participants.
- Knowledge pluralism: Elevating Indigenous, local, and experiential knowledge as essential to scientific understanding, rather than supplementary to Western science.
- Equitable partnerships: Shifting from extractive models of collaboration to long-term, trust-based relationships grounded in mutual respect and benefit-sharing.[41]
- Accessibility: Addressing and mitigating or removing the linguistic, mobility, geographic, financial, or technological barriers that prevent full participation in science and decision-making.[42,43]
- Justice and redress: Addressing entrenched systems of inequality, such as parachute science and data colonialism, and investing in structural changes in funding, authorship, and governance.

When done effectively, inclusive ocean science fosters trust, removes power imbalances, and advances ethical and equitable approaches to research that reflect and respect the lived realities of those most affected by ocean change, as well as those who have been historically marginalized.

[41] Personal conversation with Ray Santisteban Avila, Program Coordinator, Ocean World of Sound.
[42] Personal conversation with Dr. Jyotika Virmani, Executive Director, Schmidt Ocean Institute.
[43] Personal conversation with Dr. Xochitl Elías, Central America Coordinator, Early Career Ocean Professionals (ECOP) Programme.

Actionable ocean science: Represents ocean science that is timely, relevant, and usable for policy, decision-making, and community-based solutions. At its core, actionable ocean science prioritizes:

- Use-informed design: Defining research questions and outputs collaboratively and in a transdisciplinary approach with communities, decision-makers, and other end-users.
- Timeliness: Delivering results in alignment with real-world decision points, including, but not limited to, government policymaking, multilateral governance frameworks (e.g. BBNJ, CBD, UNCCD, etc.), the 2030 Sustainable Development Agenda, and the UNFCCC negotiations.
- Clarity and accessibility: Sharing outputs in multiple languages and formats to ensure they are understandable and usable by diverse audiences.[44]
- Policy and community relevance: Grounding science in local needs while ensuring scalability to broader governance and multilateral frameworks.

When done well, actionable ocean science equips a broad range of users with the science and knowledge in a timely and accessible format, supporting informed decision-making, enhanced investments, improved conservation planning, and effective policymaking. It can and must shorten the time lag between discovery and application to ensure that it is not only relevant but also addresses the urgent need to confront ocean decline.

In addition, inclusive and actionable ocean science must leverage open science and emerging technologies to maximize impact. [45] The adoption of open-access platforms, FPIC, FAIR and CARE data principles, and the responsible use of artificial intelligence, big data, and blockchain can expand accessibility, optimize the allocation of funding, and accelerate the translation of values into practice. When guided by transparency, accountability, and inclusivity, these tools enhance trust and shorten the distance between discovery, decision-making, and action.

[44] Personal conversation with Janaína Bumbeer, Fundação Grupo Boticário de Proteção à Natureza.
[45] Personal conversation with Dr. Jyotika Virmani, Schmidt Ocean Institute.

7 Co-design: A pathway to inclusive and actionable ocean science

Co-design is a platform for delivering inclusive and actionable ocean science. It aligns with the Ocean Decade's ambition to deliver transformative science that changes the current trajectory of ocean decline – 'the ocean we have' – and towards a thriving and sustainable ocean – 'the ocean we want' – by 2030.[46] Co-design shifts from science done for communities to science done with communities to create the conditions for more inclusive, legitimate, and actionable ocean knowledge. In other words, ocean science that uses the 2030 Agenda for Sustainable Development as its basis to address the most pressing issues, and then advances ocean science in a multi-stakeholder, transdisciplinary way to ensure its applicability to the solutions that are most relevant, such as policymaking, governance frameworks, conservation efforts, or technology development.[47]

Effective co-design is values-based. It is collaborative, advances diversity, equity, and inclusion, and embraces multiple knowledge systems, including natural and social sciences, as well as Indigenous and local knowledge. Importantly, co-design begins with a recognition that no single group, discipline, sector, or institution can fully address the complexity of ocean threats alone. All stakeholders bring valid perspectives, concerns, and capacities.

To guide the co-design process, the Ocean Decade promotes a three-stage model that supports the full life cycle from knowledge creation to use. The application of each phase enables actors to translate values into practice, ensuring that ocean science is both inclusive and actionable.

STAGE 1: Co-identification – initiates the process by identifying the needs of a knowledge user ('the science we need') and incorporating these into the identification of the research question and the design of the research process from the outset. That is, knowledge generators and users work together to identify the knowledge gaps and jointly frame and refine the research questions to be answered.

STAGE 2: Co-production – where, through an iterative and collaborative process involving diverse types of expertise, knowledge, and actors, research partners work to answer the identified research questions, fill knowledge gaps, and produce context-specific knowledge and pathways towards a sustainable future. The depth of engagement of knowledge users in this process can vary from matchmaking to collaboration to co-production.

STAGE 3: Co-delivery/Co-dissemination – where knowledge generators and users collectively ensure that the generated knowledge is delivered to target audiences, disseminated in a timely fashion, in user-friendly formats via accessible platforms, and is interpreted and used as intended by knowledge users, including decision makers.[48]

While traditional models of basic science remain highly valuable, co-design offers a complementary pathway that inherently fosters inclusivity, enabling a richer diversity of voices and knowledge systems, including those of Indigenous peoples, youth, and other underrepresented communities. It is not a replacement for the need for basic research, but co-designed science has demonstrated tangible benefits in creating actionable outcomes.[49]

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[46] Intergovernmental Oceanographic Commission, *Co-Designing the Science We Need for the Ocean We Want.* Paris: UNESCO, 2021.
[47] Ibid.
[48] Ibid.
[49] Personal conversation with Dr. Jason Landrum, Blue Convergence Fund.

8 The barriers to inclusive and actionable ocean science

There is a robust body of literature on the barriers to inclusive and actionable ocean science and how these impact Indigenous knowledge holders, researchers from SIDS and coastal communities, BIPOC, LGBTQIA+, early-career professionals, women, and youth. This position paper does not intend to replicate this research. Instead, it seeks to name these barriers to better highlight how the intentional application of values can move us from rhetoric to action. Importantly, many of these barriers are not the result of individual intent but rather the outcome of long-standing norms and practices that have become deeply ingrained in how science is funded, conducted, produced, disseminated, and utilized by stakeholders. The barriers discussed below can be interconnected and should not always be considered as siloed in their application to ocean science. However, if left unaddressed, they will continue to limit the reach, relevance, and impact of ocean science.

The barriers outlined are not abstract as they represent structural and systemic patterns that, when left unaddressed, limit both the inclusivity of the research process and the actionability of its outcomes. Furthermore, these barriers hinder evidence-based decision-making on policies and investments that are crucial for conserving, protecting, and sustainably utilizing marine ecosystems and resources. The table below synthesizes how each category of barrier specifically undermines one or both of these core goals, helping to clarify why a values-based transformation in ocean science is not just ethical but necessary for impact.

Table 1 - Systemic barriers to inclusive and actionable ocean science

BARRIER	WHAT IT MEANS	HOW IT UNDERMINES
Persistent power asymmetries	Ocean science remains predominantly centered in institutions in high-income countries, which control most funding, vessels, decision-making, and research priorities.[50,51,52]	Sidelines Global South scientists who need to align their efforts with the external funder.[53] Power asymmetries also impact access to funding.[54,55]
Parachute science and performative inclusion	An extractive approach where international scientists conduct research in the Global South and complete the research in their home countries, often without or just	Disconnects science from local relevance and can lead to a lack of capacity development, negatively impacting local efforts.[59]

[50] Spalding et al., "Engaging the Tropical Majority," *NPJ Ocean Sustainability*.
[51] Personal conversation with Kerstin Forsberg, Director, Planeta Oceano.
[52] Personal conversation with Elizabeth Stephenson, Director, Marine Conservation Action Fund, New England Aquarium.
[53] Personal conversation with Kerstin Forsberg, Planeta Oceano.
[54] Personal conversation with Elizabeth Stephenson, New England Aquarium.
[55] Stephenson et al., "Small grants advance global ocean conservation and management equity," *Biological Conservation.*
[59] Asha de Vos, "The Problem of 'Colonial Science'," *Scientific American*, July 2020. https://www.scientificamerican.com/article/the-problem-of-colonial-science/.

	tokenistic local participation.[56,57,58]	
Epistemic injustice	Indigenous and local knowledge systems are consistently marginalized within ocean science, often perceived as anecdotal rather than epistemologically valid or analytically rigorous.[60,61,62]	Erodes legitimacy where dominant scientific paradigms discount the credibility of knowledge holders or non-Western worldviews.[63,64]
Data colonialism and access inequities	Much of the world's ocean data is collected in the Global South but stored, analyzed, and controlled in the Global North.[65]	Local scientists and communities frequently lack access to their own data, let alone the means to analyze or apply it.[66]
Geographic and linguistic inequities	Research leadership is concentrated in English-speaking, Western institutions, where non-English work is underrecognized or distributed. Language barriers persist, creating significant barriers to participation or distribution.[67,68,69,70]	Limits the visibility and participation of non-English-speaking researchers and knowledge in ocean science.[71,72]
Exclusion from global governance	SIDS, Indigenous leaders, BIPOC, LGBTQA+, early-career scientists, women, and youth lack representation in decision-making spaces.[73]	Undermines legitimacy, access, and influence for historically marginalized groups to shape ocean policy.

[56] Paris V. Stefanoudis et al., "Turning the Tide of Parachute Science," *Current Biology* 31, no. 4 (2021): R161–R185. https://doi.org/10.1016/j.cub.2021.01.029.
[57] Personal conversation with Ken Paul, Wolastoqey First Nation.
[58] Personal conversation with Kerstin Forsberg, Planeta Oceano.
[60] Andrew Kadykalo et al., "Disentangling 'ecosystem services' and 'nature's contributions to people'," *Ecology and Society* 24, no. 4 (2019). https://doi.org/10.1080/26395916.2019.1669713.
[61] Personal conversation with Ken Paul, Wolastoqey First Nation.
[62] Cosma Cazé et al., "Co-designing marine science beyond good intentions: support stakeholders' empowerment in transformative pathways," *ICES Journal of Marine Science* 80, no. 2 (2023): 374–377.
[63] Ibid.
[64] Personal conversation with Ken Paul, Wolastoqey First Nation.
[65] Paula Marie Helm et al., "Leveraging or exploiting? Breaking through data colonialism and ethnocentrism when building a diversity-aware social platform." *AoIR Selected Papers of Internet Research 2022* (March). https://doi.org/10.5210/spir.v2022i0.13018.
[66] Ibid.
[67] Personal conversation with Janaína Bumbeer, Fundação Grupo Boticário de Proteção à Natureza.
[68] Personal conversation with Raphaël Roman, ECOP Asia Regional Coordinator.
[69] Personal conversation with Dr. Xochitl Elías, ECOP Central America Coordinator.
[70] Personal conversation with Taylor Goelz, Senior Program Manager, Ocean and Climate, The Aspen Institute.
[71] Personal conversation with Janaína Bumbeer, Fundação Grupo Boticário de Proteção à Natureza.
[72] Personal conversation with Dr. Xochitl Elías, ECOP Central America Coordinator.
[73] de Vos et al., "Towards Equity and Justice in Ocean Sciences."

Barriers to participation in global dialogues	Global South scientists, including early-career scientists, encounter significant financial and visa-related challenges in attending conferences and global events, typically held in the Global North.[74,75,76]	Skews global discourse, excluding those most impacted from participating in global convenings for career growth and capacity development.[77]
Structural exclusion from careers and leadership	Hiring practices, funding access, and social networks that disproportionately favor scientists from elite, often Western institutions.[78,79,80] Gender, race, disability, and language intersect to marginalize many researchers further.	Reinforces historical norms and practices around demographic and epistemic homogeneity in ocean science senior roles.
Inflexible academic and institutional norms	Dominant performance metrics in academia (e.g. journal impact factor, citation counts, grant income) tend to reward volume and prestige over societal relevance, inclusion, or engagement.[81]	Penalizes co-produced or community-driven research, especially for early-career researchers who see opportunities to support ocean literacy and community engagement efforts.[82,83]

[74] Personal conversation with Evgeniia Kostianaia, ECOP Programme Global Coordinator, IOC.
[75] Personal conversation with Dr. Xochitl Elías, ECOP Central America Coordinator.
[76] Personal conversation with Taylor Goelz, The Aspen Institute.
[77] Personal conversation with Evgeniia Kostianaia, ECOP Programme Global Coordinator, IOC.
[78] Rachel Bernard and Emily Cooperdock, "No Progress on Diversity in 40 Years," *Nature Geoscience* 11, no. 5 (2018): 292–295. https://doi.org/10.1038/s41561-018-0116-6.
[79] Sonya Legg et al., "Gender Equity in Oceanography," *Annual Review of Marine Science* 15 (2023): 303–329. https://doi.org/10.1146/annurev-marine-032322-100357.
[80] Personal conversation with Taylor Goelz, The Aspen Institute.
[81] Personal conversation with Dr. Xochitl Elías, ECOP Central America Coordinator.
[82] Personal conversation with Raphaël Roman, ECOP Asia Regional Coordinator.
[83] Personal conversation with Ray Santisteban Avila, Ocean World of Sound.

9 Why values matter: From barriers to action

Ocean science does not exist in a vacuum. It is shaped by human decisions and behaviors that influence who sets the priorities, what projects are funded, how data and information are collected and by whom, and which voices are supported or deprioritized.[84,85] These decisions are not neutral. They reflect values, implicit or explicit, and in the context of the Ocean Decade, they matter to ensure that science is inclusive, actionable, and responsive to the lived realities of those most affected by ocean change.

In recent years, a growing body of literature has emerged not just on the barriers to inclusive and actionable ocean science, but also calls for the centering of values to address these barriers. Much has been published on justice, equity, diversity, and inclusion (JEDI), which are key drivers of change. Similarly, pluralism powerfully expressed through *Etuaptmumk*/Two-Eyed Seeing, co-developed by Mi'kmaw Elder Dr. Albert Marshall, the late Mi'kmaw Elder Murdena Marshall, and Dr. Cheryl Bartlett, encourages the integration of both Indigenous knowledge systems and Western scientific approaches.[86] Along with others, such as accountability, humility, and transparency, these guide behavior and conduct within the science lifecycle. For funders, this is a powerful lever. Funders have direct influence on how these barriers are addressed. Certainly, every funder brings its own unique perspective and history to its work in ocean science. As such, each organization has its own set of values that guide its strategic priorities, decision-making, funding, partnerships, and activation plans. What is being asked of them now is not the adoption of new values, but the consistent, tangible, and intentional application of these values.

As noted before, this position paper does not prescribe a universal set of values that every funder must embed into its organization, as there will always be variability based on organizational focus and prioritization.[87] Further, there will always be interpretations of values based on geography and culture. Instead, this position paper highlights the values most frequently cited by actors within the ocean science ecosystem, particularly those by traditionally marginalized groups. The Foundations Dialogue acknowledges that this is not intended to be a complete list of values that any funder can include as an organizational priority. However, it serves as a starting point for further discussion on how to operationalize values in its efforts better. The Foundations Dialogue also acknowledges that many actors are conducting values-based science, and funders are applying their values to the work they support globally. The ambition is to catalyze more value-based efforts for inclusive and actionable ocean science in the second half of the Ocean Decade.

For the purposes of this position paper, there are two key categories of values for analysis:

- **Transformational values:** These values alter the structures and practices of ocean science itself.
- **Relational values:** These values shape how science is conducted in collaboration with diverse partners.

By understanding how these values influence barriers to inclusive and actionable ocean science and help build trust, funders can significantly enhance both the legitimacy and usability of ocean science, especially in complex, place-based, and rapidly changing contexts. Conversely, failing to make these

[84] Personal conversation with Dr. Rachel Clausing, Fondation Prince Albert II de Monaco.
[85] Vadrot et al., "Towards a Reflexive, Policy-Relevant and Engaged Ocean Science for the UN Decade: A Social Science Research Agenda," *Earth System Governance*.
[86] Personal conversation with Ken Paul, Wolastoqey First Nation.
[87] Personal conversation with Dr. Jason Landrum, Blue Convergence Fund.

values explicit and act upon them perpetuates dominant paradigms and reinforces exclusion under the guise of neutrality.

Table 2 - Transformational values

These values shift how science is conceptualized and practiced.

VALUE	WHAT IT MEANS	FUNDER IMPLICATIONS
Justice	Addresses historical and systemic inequities in how knowledge is produced, valued, and applied to correct the exclusion of marginalized voices, and ensures that the benefits and burdens of ocean research and governance are equitably distributed.[88,89]	Strengthens the legitimacy and effectiveness of ocean science by making it more accountable and just by addressing historic inequities and improving relationships between funders, researchers, and local communities. Promotes FAIR and CARE data principles.[90]
Equity	Ensures the fair treatment, access, and opportunity for all people, and goes beyond equal treatment to account for different starting points, structural barriers, and historic exclusion.[91,92,93]	Provides fairer access to resources, and/or leadership and career opportunities to create the conditions for success for advancement, as well as the co-design of ocean science.[94,95]
Diversity	Embraces a range of identities, disciplines, and knowledge systems, including racial, ethnic, gender, cultural, LGBTQA+, generational, geographic, transdisciplinary, and people with disabilities. Diversity alone is not enough without meaningful inclusion.[96]	Improves the quality, relevance, and applicability of ocean science that responds to knowledge holders' and users' needs, while strengthening the ocean science community through leadership, collaboration, and advancement opportunities.
Inclusivity	Creates systems, processes, structures, and environments where people with diverse identities, experiences, and	Fosters more durable science outcomes by embedding knowledge holder and knowledge user priorities into the research

[88] Erin L. Meyer-Gutbrod, James J. Pierson, and Mona Behl, "Community Perspectives on Justice, Equity, Diversity, and Inclusion in Ocean Sciences: A Town Hall Discussion," *Oceanography* 36, no. 1 (2023): 70–79. https://doi.org/10.5670/oceanog.2023.106.
[89] Personal conversation with Ken Paul, Wolastoqey First Nation.
[90] Cosma Cazé et al., "Co-designing marine science beyond good intentions: support stakeholders' empowerment in transformative pathways."
[91] Nathan J. Bennett, "Mainstreaming Equity and Justice in the Ocean," *Frontiers in Marine Science* 9 (2022): 873572. https://doi.org/10.3389/fmars.2022.873572.
[92] de Vos et al., "Towards Equity and Justice in Ocean Sciences."
[93] Written submission from the Pacific ECOP Programme, IOC.
[94] Personal conversation with Kerstin Forsberg, Planeta Oceano.
[95] Personal conversation with Elizabeth Stephenson, New England Aquarium.
[96] Gabby Ahmadia et al., "Limited Progress in Improving Gender and Geographic Representation in Coral Reef Science," *Frontiers in Marine Science* 8 (2021): 1334. https://doi.org/10.3389/fmars.2021.731037.

	knowledge systems are valued, respected, and empowered to contribute meaningfully.[97,98,99]	lifecycle to enhance legitimacy and trust in the outcomes.[100]
Pluralism	Respects and integrates multiple knowledge systems, including Indigenous and local knowledge, with Western scientific approaches, with an understanding that no single epistemology is sufficient to address complex ocean challenges.[101,102]	Elevates diverse knowledge systems to inform ocean research priorities and outcomes that can shape policy and sustainable conservation efforts to increase legitimacy and local support.[103,104]

Table 3 - Relational values

These values govern how actors work together across power, culture, and knowledge systems.

VALUE	WHAT IT MEANS	FUNDER IMPLICATIONS
Accountability	Being answerable for actions, outcomes, and decisions, especially to those most affected by them, requires openness, transparency, responsiveness, and mechanisms for redress and learning.[105]	Builds trust in institutions and actors across the ocean science ecosystem through the consistent demonstration of openness and responsiveness. Creates an opportunity to promote FAIR and CARE principles as well as FPIC.
Care	Prioritizes the physical and emotional well-being of collaborators, which is grounded in interdependence and respect.[106]	Supports a human-centered approach that fosters healthier science cultures and more sustainable partnerships and collaborations.
Humility	Recognizes the limits of one's own knowledge and the values of learning and working across	Promotes research that is grounded in listening, respect, and cultural competence that reduces epistemic injustice.[110]

[97] Personal conversation with Dr. Jyotika Virmani, Executive Director, Schmidt Ocean Institute.
[98] Spalding et al., "Engaging the Tropical Majority."
[99] Personal conversation with Ken Paul, Wolastoqey First Nation.
[100] Personal conversation with Professor Heather Koldewey, ZSL.
[101] Personal conversation with Ken Paul, Wolastoqey First Nation.
[102] Written submission from the Pacific ECOP Programme, IOC.
[103] Cazé et al., "Co-designing marine science beyond good intentions."
[104] Written submission from the Pacific ECOP Programme, IOC.
[105] Nathan J. Bennett et al., "Advancing Social Equity in and through Marine Conservation," *Frontiers in Marine Science* 8 (2021): 711538. https://doi.org/10.3389/fmars.2021.711538.
[106] Personal conversation with Elizabeth Stephenson, New England Aquarium.
[110] Ibid.

	cultures, geographies, and disciplines.[107,108,109]	
Reciprocity	Commits to mutual exchange, shared benefit, and balance between researchers and communities; thereby affirming that those who contribute knowledge, access, or time must also benefit from the outcomes.[111,112]	Reduces extractive practices and historical dynamics to build trust and promote equitable and fair collaboration across geographic, disciplinary, and generational lines.
Transparency	Makes decision-making and benefits visible and understandable to all stakeholders, including grantmaking procedures and research outcomes.[113]	Helps redistribute power by making visible who decides and when, who benefits, and how knowledge is used to encourage long-term engagement through clarity and shared expectations.

[107] Personal conversation with Ray Santisteban Avila, Ocean World of Sound.
[108] Personal conversation with Dr. Xochitl Elías, ECOP Central America Coordinator.
[109] Cosma Cazé et al., "Co-designing marine science beyond good intentions: support stakeholders' empowerment in transformative pathways."
[111] Personal conversation with Ken Paul, Wolastoqey First Nation.
[112] Personal conversation with Dr. Xochitl Elías, ECOP Central America Coordinator.
[113] Personal conversation with Taylor Goelz, The Aspen Institute.

10 Initial recommendations for funders

Navigating the complexity of systemic barriers to inclusive and actionable ocean science, as well as the numerous interconnected issues, is a challenging task. That said, as the funder community continues to support the ambitions of the Ocean Decade, there is a growing imperative to ensure that ocean science funding reflects the values outlined in this paper and enables evidence-based decision-making within the urgent policymaking and investment windows. To support these efforts, Members of the Foundations Dialogue are exploring a range of strategies, including developing a living set of guidelines for funders by the end of 2025, as well as peer-to-peer learning networks and funder collaboratives to facilitate adoption. These guidelines will provide practical, context-sensitive strategies for operationalizing values in ocean science philanthropy, from shaping calls for proposals and designing evaluation criteria to fostering long-term, co-designed partnerships.

In the interim, the Foundations Dialogue recommends four immediate actions:

1. **Conduct an internal analysis:** Map current organizational values, funding priorities, and processes to systemic barriers to understand whether your organization's approaches are aligned with desired outcomes (e.g. reducing power asymmetries, enhancing pluralism, reducing barriers to participation in global convening, etc.) to provide a benchmark and readiness assessment for ongoing planning.

2. **Evaluate current measurement systems:** Review how (or if) your organization currently tracks outputs and outcomes linked to values-based practices to understand whether, for example, co-design processes, utilization of diverse knowledge systems, or evidence uptake are being measured, which will also identify gaps where new indicators, tools, or reporting practices may be needed.

3. **Support peer learning:** Collaborate through funder-to-funder platforms, such as the Foundations Dialogue, which provide a trusted space for exchanging lessons learned, sharing tools, and exploring challenges related to grantmaking for inclusive and actionable ocean science, as well as co-design.

4. **Translate values into action:** Begin to adopt a values-based approach by piloting new processes and protocols that align with long-term organizational goals and address systemic barriers to inclusive and actionable ocean science.

These recommendations are not exhaustive, nor are they prescriptive. They provide an entry point for reflection, adaptation, and planning to create conditions where funders can be at the forefront of supporting efforts to address ocean decline and the communities most affected by it.

11 Anticipated challenges and implementation considerations for funders

Members of the Foundations Dialogue acknowledge that adopting a values-based approach to ocean science funding is a long-term process that can require internal transformation. Further, Members of the Foundations Dialogue recognize that funders must continually navigate a complex set of internal and external drivers that influence how, when, and whether values are applied to their strategic planning and decision-making.

Anticipated challenges to organizational change include:

- **Varying interpretations of values:** Even among well-intentioned institutions, the definitions of each value included in this position paper can vary, and in certain countries, these values are subject to intense scrutiny. Taken together, this can lead to the avoidance of, or the inconsistent or superficial application of, stated values.

- **Perceived risk and return on investment (ROI):** Adopting co-design and aligning funding with new values-driven approaches can be perceived as high risk, especially when working with unfamiliar partners, non-traditional knowledge systems, or place-based governance structures that do not easily map to current performance tracking metrics.

- **Board and executive approval:** Institutional change often hinges on securing support from boards or senior leadership who may be unfamiliar with the language or implications of funding co-design and a values-based approach to inclusive and actionable ocean science.

- **Foundation culture and norms:** Existing organizational cultures may prioritize legacy processes, programmes, and partnerships over deeper, community-driven initiatives, and shifting these norms requires time, trust, and consistent reinforcement of new systems, processes, and operating structures.[114,115]

- **Funding cycles and bureaucratic routines:** Rigid timelines, procurement procedures, and reporting expectations can limit the flexibility needed to support authentic co-design, relationship-building, and long-term partnerships.[116]

- **Staff capacity and expertise:** Programme officers and grants managers may lack the tools, training, or time to navigate relational grantmaking, assess inclusive practices, or engage meaningfully in co-creation processes. In particular, staff may need training on how to engage with and effectively synthesize Indigenous and local knowledge.[117]

None of these challenges is insurmountable, but they do require time, tools, and training to support institutional change. By creating spaces for reflection and experimentation, funders can reduce risk, build institutional capacity and buy-in, and accelerate the adoption of values-based ocean science.

[114] Personal conversation with Elizabeth Stephenson, New England Aquarium.
[115] Personal conversation with Taylor Goelz, The Aspen Institute.
[116] Personal conversation with Kerstin Forsberg, Planeta Oceano.
[117] Personal conversation with Ken Paul, Wolastoqey First Nation.

12 Conclusion: A values-based future for ocean science

At the midpoint of the Ocean Decade, the ocean community faces both a profound responsibility and a catalytic opportunity. The urgency of ocean decline, made clear at UNOC and the One Ocean Science Congress in June 2025, demands not only accelerated action, but transformation. Funders are not outside this system. They are embedded within it, with unique power to enable or inhibit the shifts needed to help achieve the ambitions of the 2030 Agenda for Sustainable Development and multilateral governance frameworks.

This position paper makes one thing clear: values are not new, but their consistent and intentional application is needed now more than ever. Every funder brings its own mission and priorities, yet implicit values shape the decisions they make and the actions they take. The values discussed here are not abstract ideals, but are essential for dismantling the systemic barriers that impede inclusive and actionable ocean science. It is intended to serve as a roadmap for other funders.

In the coming months, the Foundations Dialogue will:

- Convene funders of ocean science for further discussions on how to operationalize values to advance inclusive and actionable ocean science.
- Develop a set of guidelines to demonstrate how transformational and relational values can be incorporated in organizational planning, grantmaking, and relationship-building.
- Design a measurement framework to track how values-based practices translate into outputs and outcomes, including standardized indicators (e.g. co-designed reports, funding calls shaped by multiple knowledge systems, inclusion metrics, evidence-based policy uptake, knowledge-sharing).
- Initiate peer-to-peer learning forums to share case studies, practical tools, and lessons learned on co-design and values-based ocean science.
- Pilot funding practices and collaborative initiatives that embed co-design to demonstrate the tangible benefits of inclusive approaches and generate actionable science.

This is not a call for perfection. It is a call for courageous action. Funders will face tensions and tradeoffs across governance, grantmaking, institutional culture, and accountability, but these are not insurmountable. Through tools like co-design, values-based funding practices, peer-to-peer learning, and shared platforms for transparency and collaboration, change is already underway. The Foundations Dialogue and its members are committed to leading by example and continuing to learn from those most affected by ocean change.

If the next five years of the Ocean Decade are to be defined by impact rather than inertia, then funders must also move from rhetoric to action. The result will enhance our collective ability to deliver a more inclusive and equitable ocean science ecosystem, providing timely and trusted knowledge and information to inform policymaking and conservation efforts. And this is how we achieve "the science we need for the ocean we want."

Appendix – Values in action: Case examples

Already, funders are demonstrating what they have learned to address barriers to inclusive and actionable ocean science. Across the global ocean community, funders have begun to integrate co-design, elevate underrepresented leadership, and dismantle barriers to systemic exclusion. Crucially, many of these efforts embrace not only successes but also "successful failures," where projects or grants may not have gone as planned but yielded valuable insights into what alternative approaches are necessary to foster inclusive and equitable pathways to deliver timely, usable, and relevant knowledge.[118] The following case examples highlight initiatives where values have been explicitly and intentionally applied to ocean science challenges. These are not isolated success stories, but tangible evidence that the philanthropic community is advancing from rhetoric to practice and the benefits that are being generated. They offer diverse models for how funders can act on transformational and relational values in real-world contexts.

Belmont Forum: OCEAN 2: Toward the Ocean We Want

The Belmont Forum's "OCEAN 2: Towards the Ocean We Want" Collaborative Research Action responds to the urgent need for transdisciplinary, solutions-oriented research that addresses accelerating threats to ocean biodiversity and ecosystem sustainability. Fragmented scientific efforts and the underrepresentation of non-academic voices have historically limited the uptake of ocean knowledge in policy and practice. To overcome these gaps, the initiative emphasizes co-creation, co-design, and co-implementation within multinational consortia that integrate academic researchers from the natural and social sciences with societal partners (e.g. policymakers, non-governmental organizations, Indigenous peoples, private sector actors, and local communities) from the outset. It explicitly targets three thematic areas: (1) Biodiversity conservation and nature-based solutions; (2) Ocean-biodiversity-climate integration; and (3) Nature futures, ocean governance, and ethics for sustainability. Projects may be submitted in English, French, Portuguese, or Spanish, thereby fostering greater inclusion and accessibility, particularly for researchers and communities from the Global South. The call is supported by research agencies in Canada, Côte d'Ivoire, France, Italy, Norway, Switzerland, Thailand, Uruguay, and others, alongside philanthropic partners such as the Schmidt Ocean Institute. Through its flexible blended funding mechanism, the Belmont Forum promotes actionable science and fosters approaches that enable societal partners to access direct financial support. This balances science and society, turning knowledge into action through meaningful participation at every stage of a project, from design to implementation, and embodies the Belmont Forum's transformational and relational values in overcoming systemic barriers. Aligned with the Ocean Decade, OCEAN 2 reinforces commitments to open data policies and advances science that generates robust evidence to support international agreements.

Great Barrier Reef Foundation

The Great Barrier Reef Foundation is addressing the urgent need to build ocean solutions that center the knowledge, leadership, and economic inclusion of Indigenous communities, especially those living on the frontlines of climate and biodiversity change. Recognizing that Indigenous Peoples comprise just 5% of the global population yet steward over 80% of the planet's remaining biodiversity, the Foundation is investing AUD$25 million to co-develop a nature-based economy in partnership with Traditional Owners of the Great Barrier Reef. This initiative is designed by Traditional Owners for Traditional Owners and seeks to establish culturally grounded, community-led enterprises that simultaneously support environmental stewardship and inclusive economic development. Through

[118] Personal conversation with Professor Heather Koldewey, ZSL.

scalable business models, strong cultural brand identities, and expanded industry partnerships, the programme empowers reef communities to care for the country in ways that reflect their deep knowledge and custodianship. This work exemplifies the Ocean Decade principles of co-design, sustainable ocean-based economies, and inclusive governance, offering a practical model for aligning ocean conservation with social and economic justice.

Early Career Ocean Professionals (ECOP) Programme: Ocean Decade

The ECOP Programme, a flagship initiative of the Ocean Decade, empowers a new generation of ocean professionals by creating a global polycentric framework for capacity development, leadership, and inclusive participation. Recognizing that early-career professionals, those with fewer than 10 years of experience across all sectors related to the ocean, bring fresh perspectives and urgency to ocean sustainability, the Programme facilitates meaningful engagement through mentorship, training, and participation in governance processes. Since its launch in 2021, the ECOP Programme has grown to over 7,000 members, elevating the collective voice of early-career professionals across geographies and disciplines. Its goal is not only to support professional growth but also to embed their knowledge and leadership into Decade-wide actions, with a strong emphasis on equity, inclusion, and intergenerational knowledge transfer.

Fondation Prince Albert II de Monaco

In Arctic regions, Indigenous voices and knowledge systems remain under-represented in many scientific conversations and policy processes.[119] The Fondation Prince Albert II de Monaco (FPA2)'s Polar Initiative (launched in 2022) aims to strengthen Indigenous participation, recognize Indigenous knowledge systems, and support equitable access to decision-making in the Arctic marine sphere, linking science, policy, and Indigenous/local governance to address the rapidly changing Arctic environment. The FPA2's grantmaking process obliges co-design and community management: no projects on Indigenous issues are supported unless the communities are directly co-designing and implementing them, reflecting a shift from consultation to genuine partnership and empowerment. Funding of intermediaries is also avoided. For example, the FPA2-supported project to reinforce capacity of Sámi communities in decision-making processes funds the Saami Council directly (€60,000, 2026). Support is also designated for inclusivity and preservation of traditional knowledge and languages, such as a series of projects with the Qikiqtani Inuit Association (2012, 2019, 2024) that ensured translation of scientific and policy documents into Inuktitut and Inuvialuktun and production of a series of atlases containing knowledge of Inuit elders in Inuktitut dialects (eight distinct communities, >€100,000).

As funding to the blue economy grows alongside concerns about transparency and equity, the FPA2 embeds accountability and transparency at the core of its blue economy investments. Its ReOcean Fund – co-founded with Monaco Asset Management and structured as a Luxembourg Reserved Alternative Investment Fund – is a late-stage venture capital/growth impact fund under Article 9 of the Sustainable Finance Disclosure Regulation, designed to channel capital towards an inclusive and transparent blue economy. It supports ventures that integrate social equity, climate resilience, and ocean health across five verticals – from blue food to restoration and data – governed by independent impact oversight and aligned with SDGs 7, 12, 13, 14, and 15. The FPA2's philosophy to empower local leadership and ensure transparent metrics for measurable impact also inspired its co-founding and continued support of the Global Fund for Coral Reefs (GFCR), a blended-finance vehicle targeting more than 20 developing coral reef nations to build durable, scalable solutions.[120] The GFCR Stakeholder Engagement Policy enshrines inclusive and transparent participation by communities,

[119] Joy M. O'Brien et al., "Ten 'simple' rules for non-Indigenous researchers engaging Indigenous communities in Arctic research," *PLOS Computational Biology* 20, no. 6 (2024): e1012093. https://doi.org/10.1371/journal.pcbi.1012093.
[120] UNESCO, "UNESCO recognizes the Global Fund for Coral Reefs' critical contribution to the UN Decade of Ocean Science" (2023). https://www.unesco.org/en/articles/unesco-recognizes-global-fund-coral-reefs-critical-contribution-un-decade-ocean-science.

Indigenous peoples, women, and other marginalized groups throughout project design and implementation, aligning with global best practices to ensure social and environmental sustainability.[121,122] Together, these efforts embody the Ocean Decade vision for sustainable ocean-based economies and demonstrate the FPA2's conviction that co-investment and inclusive systems can deliver lasting benefits for both people and planet.

Lenfest Ocean Program: Supporting inclusion of Indigenous knowledge

The Lenfest Ocean Program (LOP) at The Pew Charitable Trusts recently sunsetted after 20 years of impact, supporting useful and usable science positioned to inform marine policy and decision-making through the co-design and co-production of knowledge. During this time, the Program evolved its grantmaking approach to put a greater focus on community-informed science needs, and associatively the integration of the values of diversity, equity, inclusion, and transparency in their grantmaking and outreach efforts. One of the outcomes of this shift resulted in a portfolio of projects developed to elevate the inclusion of Indigenous knowledge (IK) in ocean and coastal decision-making in the United States and Canada. There was a clear need to support these communities and their knowledge systems: historically, the traditional knowledge of Indigenous Peoples has been overlooked, or simply disregarded, in decision-making processes, impeding the effective conservation and restoration of marine ecosystems. But, as a funder, it was important to be intentional and proactive in ensuring that the grantmaking process was inclusive, equitable, and accessible to Indigenous communities who would ultimately lead the projects. Some of the key actions to achieve this goal are mentioned below.

First, to be broadly accessible to as many communities as possible, the funding opportunity was announced and advertised as an open call for proposals that specifically prioritized supporting Indigenous institutions and communities directly. Second, to minimize unconscious bias in the grantmaking process, Indigenous experts were invited to the grantmaking process at several points: reviewing the request for proposals' language to make sure that it was inclusive, co-designing the selection criteria used to review submissions, and contributing as key members of a review panel that selected top applicants considered for funding. Third, LOP staff worked with other Pew colleagues to facilitate fairer terms for the project teams and the communities for which they were working, including greater flexibility to fund support staff and operations, and a more prominent role in securing, safeguarding, and applying the IK associated with the proposed work (i.e. data sovereignty). Collectively, these actions were deliberately and systematically included to foster a more trust-based approach to the funder-grantee relationship, integrate more equitable practice in overall grantmaking processes, and ensure more direct "power" transfer between the funder and its project partners.

One additional point to emphasize in the context of this current paper on values that guide funders' support and engagement: the decision to develop and implement the IK portfolio described above was not done without prior thought and planning by LOP. Ahead of developing the IK portfolio, the staff took a journey together to explore the Program's values, defining the terms diversity, equity, inclusion, and transparency (DEIT) in relation to the initiative's principles and practices. Unpacking the unconscious biases held by the team members led to a formalization of these values and definitions that guided priority setting and implementation of actions.

[121] Global Fund for Coral Reefs, *Stakeholder Engagement Policy* (2023). https://globalfundcoralreefs.org/reef-plus/knowledge/gfcr-stakeholder-engagement-policy/.
[122] Global Fund for Coral Reefs, *Policies Hub* (2023). https://globalfundcoralreefs.org/reef-plus/knowledge/global-fund-for-coral-reefs-policies-hub/.

Marine Conservation Action Fund (MCAF): New England Aquarium

The New England Aquarium's Marine Conservation Action Fund (MCAF) supports grassroots ocean conservation led by local experts in low- and middle-income countries. Through small grants and fellowships, MCAF enables community-based leaders to advance projects that merge ocean health with social well-being. MCAF operates on long-term partnerships, trust-building, and mutual learning, fostering a global community of practice grounded in equity and care. The fund's bottom-up approach challenges conventional funding cycles by investing in people first, prioritizing relationships, capacity-building, speed to funding, and co-creation over transactional grantmaking. This approach has enabled conservation leaders to create culturally relevant, scalable solutions while strengthening the leadership and legitimacy of Global South practitioners in ocean science and governance.

REV Ocean and IOC: Ocean Matcher

REV Ocean and IOC launched Ocean Matcher, an AI-powered digital platform designed to address one of the most persistent barriers to inclusive and actionable ocean science: the misalignment and delay of funding. Built in partnership with NoA Ignite and supported by a coalition of philanthropic foundations, Ocean Matcher uses natural language processing and algorithmic matchmaking to connect ocean scientists, innovators, and project leaders with funders who share aligned values and priorities. By streamlining the identification and matching of high-impact proposals with capital, the platform supports co-design and co-financing, reduces inefficiencies, and enhances equitable access to funding. Ocean Matcher is the result of two years of co-development, beginning with early discussions at the 2023 Foundations Dialogue in Monaco, and represents a bold, values-driven effort to accelerate the delivery of ocean solutions. By enabling transparent, timely, and targeted funding, it reflects the Ocean Decade's call for technology and philanthropy to work together in transforming ocean science from knowledge to action.

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